

LEVEL 1 — TEST 1: BASELINE STRESS TEST

SEWAC Multi-Vehicle Telemetry Architecture

Abstract

This report presents the results of Level 1, Test 1 (Baseline) of the SEWAC multi-vehicle telemetry architecture. The objective was to evaluate concurrent telemetry generation and transmission from three simulated vehicles operating across Ward 1, Ward 2, and Ward 20. The simulation ran for 10 minutes with independently randomized packet intervals of 5–12 seconds. A total of 200 telemetry packets were generated and all 200 were successfully transmitted, resulting in a 100% sending success rate. However, only 63 packets were observed at the receiving end, corresponding to an observed receiving rate of 31.5% and 137 packets not observed downstream. Three apparent bottlenecks were observed in the receiving/buffering path. The test therefore constitutes a partial pass and establishes the receiving/buffering layer as the primary area requiring investigation before higher stress levels are applied.

Index Terms

SEWAC, telemetry, multi-vehicle simulation, stress testing, buffering, receiving pipeline, baseline evaluation, IoT telemetry.

I. Introduction

The SEWAC telemetry architecture is designed to receive telemetry packets generated by multiple vehicles operating simultaneously within a hierarchical geographic structure consisting of cities, zones, divisions, and wards. Before increasing system load, a baseline test was conducted to establish the initial performance characteristics of the telemetry path.

Level 1, Test 1 was configured as a low-intensity baseline scenario. Three vehicles transmitted telemetry independently using randomized intervals rather than a synchronized schedule. This configuration was intended to determine whether the architecture could reliably handle concurrent, non-periodic telemetry traffic under relatively low load.

II. Test Architecture and Configuration

The baseline simulation consisted of three vehicles distributed across the configured test wards. Each vehicle generated telemetry packets independently and transmitted them throughout the ten-minute test window.

Vehicle	Assigned Ward	Telemetry Unit
KA05AB1234	Ward 1	SEWAC_01_UHF
KA05AB1235	Ward 2	SEWAC_01_UHF
KA05AB1236	Ward 20	SEWAC_01_UHF

The test duration was 10 minutes. Packet transmission intervals were randomized independently for each vehicle between 5 and 12 seconds. This prevented artificial synchronization between vehicle streams and provided a concurrent telemetry workload.

III. Test Objective

The primary objective was to establish a baseline for the multi-vehicle telemetry pipeline and determine whether packets generated by three concurrent vehicle streams could be transmitted and subsequently received without loss. Particular attention was given to the distinction between successful packet transmission and successful downstream reception, since these represent different stages of the architecture.

IV. Experimental Procedure

All three vehicle simulations were started concurrently. Each simulation independently selected telemetry records and generated packets at randomized intervals within the configured 5–12 second range. Packets were transmitted for the complete 10-minute duration. The sending-side simulator recorded transmission outcomes, while the receiving-side observation was used to determine how many transmitted packets were subsequently observed.

No intentional packet-generation or transmission failures were introduced during Test 1. The purpose of this level was to establish the normal operating baseline before introducing progressively higher traffic intensity.

V. Results

A. Overall Transmission Results

Metric	Result
Test Level	Level 1
Test Number	1
Test Name	BASELINE
Duration	10 minutes
Random Packet Interval	5–12 seconds
Total Packets Sent	200
Successful Transmissions	200
Transmission Failures	0
Transmission Success Rate	100%

The sending-side results demonstrate complete success during the baseline test. All 200 generated packets were successfully transmitted, and no transmission failures were recorded. This establishes that the vehicle simulation and packet submission mechanism were stable under the Level 1 workload.

B. Receiving-End Results

Vehicle	Ward	Packets Actually Received
KA05AB1234	Ward 1	12
KA05AB1235	Ward 2	21
KA05AB1236	Ward 20	30
Total	—	63

Although 200 packets were successfully transmitted, only 63 packets were observed at the receiving end. The difference between transmitted and observed packets is therefore 137 packets. Based on the supplied receiving-end observation, the observed receiving rate was 31.5%, while the apparent unobserved or missed-packet rate was 68.5%.

C. Quantitative Performance Analysis

The receiving rate was calculated as $63/200 \times 100 = 31.5\%$. The corresponding shortfall was $200 - 63 = 137$ packets, representing $137/200 \times 100 = 68.5\%$ of the transmitted packet volume.

VI. Observed Bottlenecks

The principal observation from Test 1 was a discrepancy between the sending and receiving stages. The sending side reported 100% successful transmission, whereas the receiving side observed only 63 packets. Three apparent bottlenecks were identified in the receiving/buffering path during the test. These bottlenecks caused a substantial number of packets to be missed or not observed downstream.

The results indicate that the primary limitation at this test level is not packet generation or initial transmission. Instead, the receiving and buffering path requires further examination. The exact internal cause of the three bottlenecks was not isolated by this test alone and should therefore be investigated through additional instrumentation and queue/buffer monitoring.

VII. Discussion

The baseline experiment successfully separated the performance of the transmission stage from that of the receiving stage. The perfect sending result demonstrates that the simulation was capable of producing concurrent randomized traffic without transmission-side failure. This confirms that the generated telemetry workload reached the initial system boundary as intended.

Conversely, the receiving result demonstrates that successful transmission does not guarantee successful downstream retention or processing. The difference of 137 packets indicates a substantial capacity or buffering issue somewhere after the initial transmission stage. The unequal distribution of received packets across the three vehicle streams further indicates that receiving behavior was not uniform across the concurrent streams.

Because this was a baseline test, the observed bottleneck should be treated as a critical architectural finding rather than as evidence that the complete architecture is incapable of handling higher loads. Higher stress levels should be conducted only after the receiving path has been instrumented sufficiently to identify where packets are being delayed, overwritten, dropped, or otherwise not observed.

VIII. Limitations

The receiving-end measurements supplied for this report represent packets actually observed at the receiving end and do not, by themselves, identify the exact point at which the remaining packets were lost or missed. Consequently, the reported 68.5% figure should be interpreted as an observed receiving-side shortfall rather than a definitive attribution of packet loss to a specific software component. Additional queue-depth, consumer, buffer, and persistence measurements are required for precise root-cause identification.

IX. Conclusion

Level 1, Test 1 established the baseline performance of the SEWAC multi-vehicle telemetry simulation. During the ten-minute experiment, 200 telemetry packets were successfully generated and transmitted, producing a 100% transmission success rate. The receiving side, however, observed only 63 packets, corresponding to an observed receiving rate of 31.5% and a shortfall of 137 packets. Three apparent bottlenecks were identified in the receiving/buffering path.

Accordingly, Level 1, Test 1 is classified as a PARTIAL PASS. The vehicle simulation and transmission pipeline passed the baseline requirement, while the receiving/buffering layer requires investigation and optimization. The results provide a clear baseline against which subsequent stress levels can be compared.

X. Recommended Next Action

Before proceeding to higher stress levels, the receiving and buffering pipeline should be instrumented to record queue depth, enqueue count, dequeue count, consumer throughput, processing latency, and packet disposition. Once these measurements are available, the next test level can determine whether the receiving bottleneck scales predictably with increased traffic or whether a specific buffering or consumer constraint is responsible for the observed shortfall.

Table I — Final Level 1 Test Summary

Parameter	Observed Result
Test	Level 1 — Test 1
Scenario	Baseline
Duration	10 minutes
Vehicles	3

Packets Sent	200
Transmission Success	200 / 200 (100%)
Packets Received	63
Packets Not Observed	137
Observed Receiving Rate	31.5%
Apparent Receiving Shortfall	68.5%
Receiving Bottlenecks	3
Overall Assessment	PARTIAL PASS